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Metal-coordinated Phthalocyanines as Platform Molecules for Understanding Isolated Metal Sites in Electrochemical Reduction of CO₂

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ABSTRACT: Single-atom catalysts (SACs) of non-precious transition metals (TMs) often show unique electrochemical performance, including for the electrochemical carbon dioxide reduction reaction (CO₂RR). However, the inhomogeneity in their structures makes it difficult to directly compare SAC of different TM for their CO₂RR activity, selectivity and reaction mechanisms. In this study, the comparison of isolated TM (Fe, Co, Ni, Cu and Zn) is systematically investigated using a series of crystalline molecular catalysts, namely TM-coordinated phthalocyanines (TM-Pcs), to directly compare the intrinsic role of the TMs with identical local coordination environments on the CO₂RR performance. The combined experimental measurements, *in-situ* characterization, and density functional theory calculations of TM-Pc catalysts reveal a TM-dependent CO₂RR activity and selectivity, with the free energy difference of $\Delta G(^*HOCO) - \Delta G(^*CO)$ being identified as a descriptor for predicting the CO₂RR performance.

1. Introduction

While precious metals such as gold (Au),¹⁻³ silver (Ag),^{4,5} and palladium (Pd)⁶⁻⁸-based electrocatalysts have been extensively investigated for CO₂RR, single-atom catalysts (SACs) have also attracted significant attention because of the demand for developing cost-effective catalysts.⁹⁻¹⁸ Despite the high activity and cost-effective features of SACs of transition metals (TMs) originating from the optimal metal utilization, the catalytic mechanism and the local environment surrounding TM centers have not been clearly understood because of the different defect moiety (N, S, and O, *etc.*) in carbonaceous supports that can stabilize TM and lead to different electronic/geometric effects.¹⁹⁻²² The different CO₂RR activities observed among SAC of different TMs, or even in the same SAC prepared with different synthetic conditions¹⁸ (*e.g.*, annealing temperatures), also suggest the ambiguity in the SAC structures and their resultant activities. For example, drastically different CO selectivity was reported for Co-SACs¹³⁻¹⁷ (94 % vs. 24 %), thus highlighting the urgency and significance to directly compare SACs of different TMs with identical local coordination environments to understand the origin of their distinct CO₂RR performance.

In this study, we systematically investigate the intrinsic effect of an isolated TM environment on CO₂RR by using a series of TM-coordinated phthalocyanines (TM-Pcs, TM=Fe, Co, Ni, Cu, and Zn), where the well-defined metal-nitrogen (TM-N₄) units are present for all the TM-Pcs. The

CO₂RR product distribution was significantly affected by the choice of different TM. At -0.7 V versus reversible hydrogen electrode (RHE), CO selectivity was in the order of Co-Pc > Ni-Pc >> Fe-Pc > Cu-Pc > Zn-Pc, and the CO partial current density of Co-Pc was nearly 3 orders of magnitude higher than that of Zn-Pc, which was different from the recent reports on SACs where Co-based SAC¹³⁻¹⁵ was exclusively active for the hydrogen evolution reaction (HER), a main competing reaction against CO₂RR. *In-situ* X-ray characterization measurements confirmed that the TM-N₄ coordination of isolated TMs was retained during CO₂RR. Density functional theory (DFT) calculations identified a general descriptor (free energy difference between ^{*}HOCO and ^{*}CO) to understand and predict the CO₂RR performance of different TM-Pcs, with a linear correlation between theoretical DFT calculations and experimental data over the entire potential range in this work. The findings in the current investigation should provide insights into understanding the role of different isolated TM and a general descriptor to help design improved CO₂RR catalysts.

2. Results and Discussion

2.1. Catalyst Characterization

TM-Pc is a molecular crystalline material consisting of divalent TM and the planar Pc²⁻ ligand (**Figure 1A**) under the D_{4h} (square planar) symmetry group.^{10, 23, 24} The crystalline characteristics of TM-Pcs were confirmed by synchrotron-based X-ray diffraction (XRD) analysis with a wavelength of 0.24117 Å (**Figure 1B**). The Le bail fit of the XRD patterns revealed that TM-Pc materials hold either monoclinic β -phase²⁵⁻²⁷ ($P2_1/c$) or triclinic α -phase²⁸ ($P-1$). The fitting results are shown in **Figure S1** and **Table S1**. Except for Zn-Pc, the TM-Pcs exhibited needle or rod like morphology in scanning electron microscopy (SEM) characterization (**Figure S2**), which is consistent with previous studies.²⁹⁻³³ The D_{4h} symmetry around TM was confirmed by X-ray absorption fine structure (XAFS) analysis (**Figure 1C** and **Figure S3**). In the X-ray absorption near edge structure (XANES) profile, the negligible intensity of the pre-edge peak (Peak A) of TM-Pcs indicated the centrosymmetry feature of the D_{4h} symmetry. In addition, all of the TM-Pcs showed a noticeable intensity of Peak B, corresponding to the $1s$ -to- $4p_z$ transition, supporting the TM environment coordinated with four nitrogen atoms (*i.e.*, TM-N₄) in the D_{4h} symmetry.^{23, 24} One noticeable peak around ~ 1.5 Å in the extended-XAFS (EXAFS) profiles of TM-Pcs (**Figure S4**), corresponding to the TM-N bond, was consistent with the characteristics of isolated TM surrounded by four nitrogen atoms (**Table S2** and **S3**). EXAFS fitting also indicated that TM is coordinated by 3.8~4.3 nitrogen atoms with the bond length ranging from 1.89~1.99 Å, further supporting the well-defined TM-N₄ unit in the TM-Pc molecules. The overall structural analyses indicate that all of the catalysts have a well-defined isolated-TM²⁺ environment with four nitrogen atoms in the conjugate Pc unit without the structural inhomogeneity^{12, 14, 16} observed in the TM-based SACs, typically referred to as TM-N_x ($x=1\sim 4$).

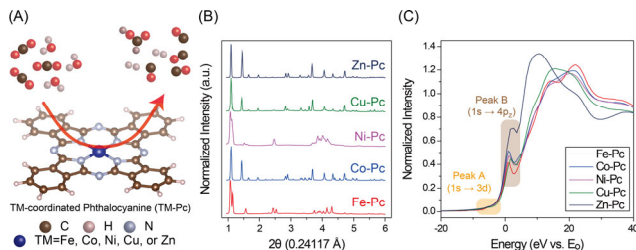


Figure 1. Structural analysis of TM-Pcs. (A) Schematic illustration for CO₂RR over a chemical unit of the TM-Pc. (B) The synchrotron-based XRD profiles of a series of TM-Pc. (C) XANES profiles of powder TM-Pcs at different TM K-edges.

2.2. Electrochemical Measurements

The electrochemical CO₂RR activity of a series of TM-Pcs was evaluated using the chronoamperometry method at different potentials ranging from -0.6 V to -1.1 V versus RHE in CO₂-saturated 0.5M potassium bicarbonate

(KHCO₃) aqueous solution (See Methods for the details, **Figure S5**). All potentials (V) are referenced to RHE in the current study. It is found that all TM-Pcs produce syngas (CO/H₂) as their main products with formic acid (HCOOH) as a minor liquid product, as evidenced by the Faradaic efficiency (FE) profiles of the products (See Methods and Supporting Information for the detailed FE calculation, **Figure 2**), thus allowing us to use syngas production to compare CO₂RR activity on different TM-Pcs.

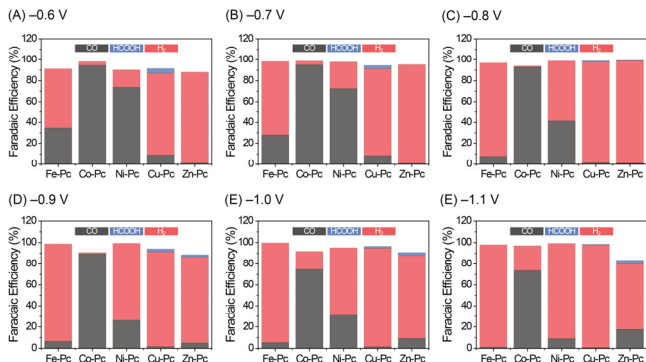


Figure 2. Faradaic efficiency of different products for a series of TM-Pcs at (A) -0.6 V, (B) -0.7 V, (C) -0.8 V, (D) -0.9 V, (E) -1.0 V, and (F) -1.1 V.

The FE profiles for H₂, CO, and HCOOH over TM-Pcs at -0.7 V are shown in **Figure 3A**. At this potential, CO and H₂ are the main products compared to HCOOH. It is clear that CO production is apparently influenced by the choice of the TM center in the order of Co-Pc (highest, 95 %) > Ni-Pc (~ 72 %) >> Fe-Pc (~ 28 %) > Cu-Pc (~ 8.5 %) > Zn-Pc (lowest, 0.8%). These results imply that the TM center plays a role in modifying the binding energies of key intermediates involved in both CO₂RR and HER, which will be discussed later in the DFT calculations. Especially, Co-Pc shows the better FE(CO) than the other TM-Pcs throughout the entire potential range and maintains the high FE(CO) of 73.7 % at -1.1 V (**Figure 2E**).

The partial current density of CO ($j(\text{CO})$, See Experimental Methods in Supporting Information for the detailed calculation) also shows a similar TM-dependent profile to the FE trend with Co-Pc being most active for CO production (~ 8.6 mA cm⁻² at -0.7 V), significantly higher than the other TM-Pcs at the same condition (**Figure 3B** and **Figure S6**). The mixed catalyst consisting of Co-Pc (most active) and Zn-Pc (least active) in a weight ratio of 50:50 exhibits ~ 3.15 mA cm⁻² of $j(\text{CO})$ and 78 % of FE(CO) at -0.7 V (**Figure S7**), which provides additional opportunities of controlling the CO/H₂ ratio using mixtures of TM-Pc based catalysts. The syngas (CO/H₂) ratio also indicates that Co-Pc is efficient at selectively converting CO₂ into CO as shown in **Figure 3C** and **Figure S8**. The FE(CO) and $j(\text{CO})$ values of the catalysts in this study are tabulated in **Table S4** in comparison with those of similar TM-Pc-based catalysts.³⁴⁻⁴¹ It is important to point out that the primary

focus of the current study is to compare the CO₂RR performance of different TM-Pc molecules under identical electrochemical conditions to determine the intrinsic role of isolated TM. Considering that the TM-Pc structure has been often employed as a structural model of TM-SACs due to the potential presence of TM-N₄ complex in the TM-SACs,^{11-13, 15-18, 42-44} the selective CO production of Co-Pc should be counterintuitive against the previous works.¹³⁻¹⁵ This is because Co-based SACs have been reported to be exclusively active for HER rather than CO₂RR. Therefore, the electrochemical results in this study imply that the Co-N₄ configuration should be active for CO₂RR rather than HER, and the HER preference in Co-SACs most likely originates from the different isolated Co environment with that in the Co-Pc as will be discussed in the DFT section. **Figures 3D and 3E** show the stable electrochemical performance of Co-Pc and Zn-Pc for 3 h. The selective CO production (FE(CO)=92~95 %) is maintained for Co-Pc even after 12 h of CO₂ electrolysis (**Figure S9**) while Zn-Pc retains its preference for HER (FE(CO)=1~2 %), confirming the robust TM-N₄ moiety under CO₂RR conditions. It should be pointed out that the conductivity of electrocatalysts also plays a potential role in controlling the electrocatalytic performance.^{45, 46} The electric conductivity of the TM-Pc/C catalysts should be further evaluated and optimized for their potential applications in electrochemical devices.

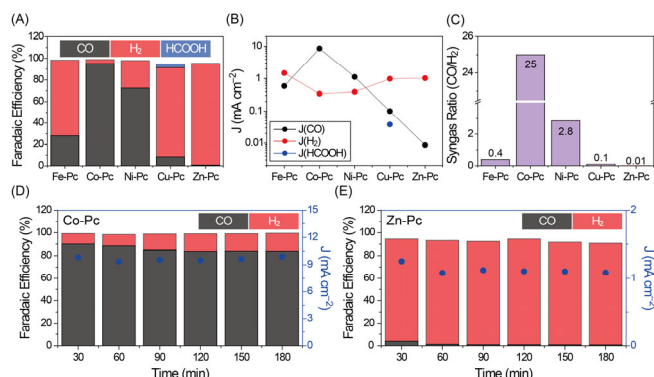


Figure 3. Electrochemical CO₂RR performance of TM-Pcs at -0.7 V. (A) Faradaic efficiency of each product, (B) Partial current density of each product, and (C) Calculated CO/H₂ ratios of TM-Pcs. Stability tests of (D) Co-Pc and (E) Zn-Pc.

2.3. In-situ Characterization

In order to verify if the physicochemical properties of TM-Pcs were modified during CO₂RR, *in-situ* X-ray characterization was carried out. See Experimental Methods in Supporting Information for the details. *In-situ* XRD analysis was first conducted to monitor the structural changes of TM-Pcs during the CO₂RR (**Figure 4A-E**). However, negligible changes were observed in the XRD patterns, confirming that all of the TM-Pcs maintained their original frameworks at the bulk scale. Furthermore, XRD patterns

at higher 2 θ ranges also remained the same, thus supporting that both long- and short-range ordering in TM-Pcs was not altered during the CO₂RR (**Figure 4F** and **Figure S10**). This is also consistent with the high-resolution electron microscopy observation that elemental distribution remained the same before and after CO₂RR at -0.7 V (**Figure S11** and **Figure S12**).

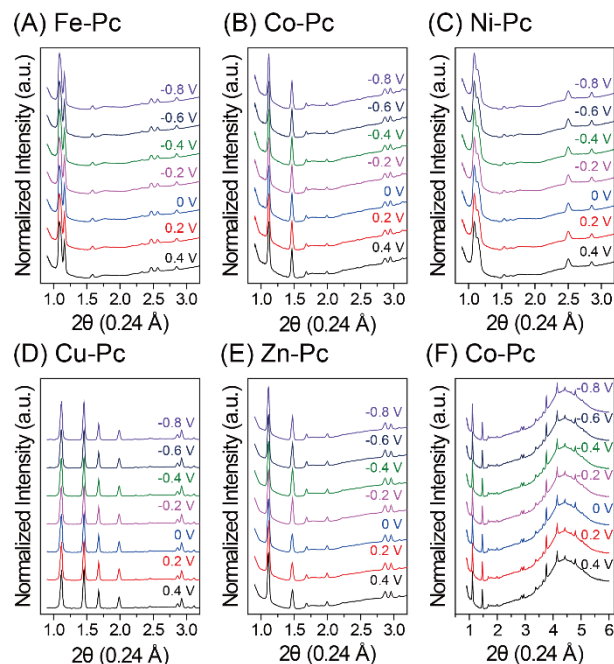


Figure 4. *In-situ* XRD analyses for a series of TM-Pcs in the range from 0.4 V to -0.8 V with an interval of 0.2 V.

In-situ XAFS analysis also revealed the robust TM-N₄ structure during the CO₂RR. TM-Pcs showed negligible changes in both XANES (**Figure 5**) and EXAFS (**Figure S13**) profiles throughout the entire potential range due to the strong covalent interaction between elements in TM-Pc molecules. In particular, as shown in the near edge energy region (**Figure 5A-E**), the profiles did not change during CO₂RR, supporting that the local coordination of TMs (*i.e.*, TM-N₄) in the D_{4h} symmetry group was retained. Furthermore, in the extended energy region, the peaks, corresponding to the multiple scattering process of photoelectrons in the lattices, did not shift, indicating that the bivalent TM nature in TM-Pc molecules was maintained (**Figure 5F** and **Figure S14**). Furthermore, the absence of any change in the TM-N bond length, corresponding to the peak around ~1.5 Å, confirmed again the stable TM-Pc structure (**Figure S15**) as revealed in *in-situ* XRD analyses. Such durability in the catalyst structure is also reflected by the stable electrocatalytic performance as shown in **Figure 3D and 3E**.

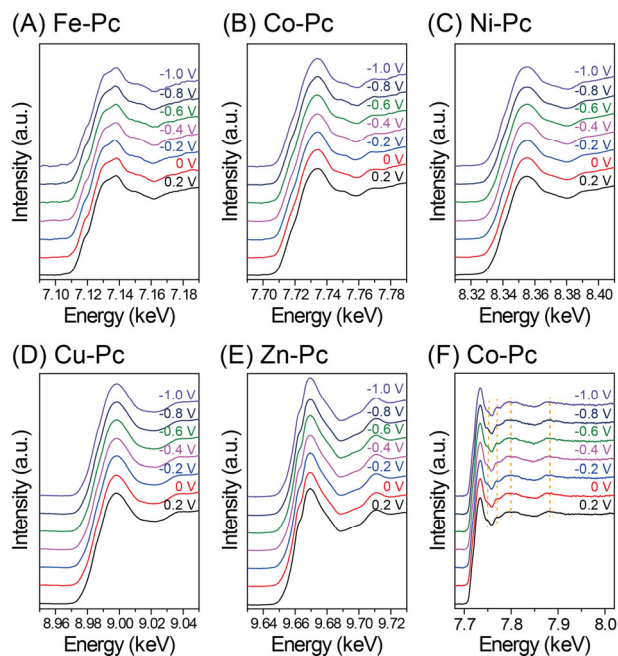


Figure 5. *In-situ* XANES analyses for a series of TM-Pcs in the potential range from 0.2 V to -0.8 V with an interval of 0.2 V.

2.4. DFT Calculations

DFT calculations^{2, 47-54} were performed to gain insights into the varied CO₂RR activity of different TM-Pc electrocatalysts. Detailed computational methods are described in Supporting Information. The binding energies of CO₂RR intermediates (*i.e.*, *HOCO and *CO) and HER intermediate (*i.e.*, *H) were calculated in their lowest energy adsorption configurations (**Figure S16 and Figure S17**). The results show a significant variation in calculated binding energies between various TM-Pcs (**Table S5**). In **Figure 6A**, the free energy diagrams for CO₂RR, computed based on the DFT calculated binding energies using the computational hydrogen electrode model developed by Norskov and coworkers,⁵⁴ suggest that Co-Pc is predicted to show the best CO₂RR performance among all TM-Pcs because of the largest stabilization of *HOCO and moderate *CO binding making its desorption a facile process, which is in agreement with the trend of FE(CO) and J(CO) measured in experiments (**Figure 3A-C**).

Furthermore, as shown in **Figure 6B**, a linear correlation between $\Delta G(^*H)$ and $\Delta G(^*HOCO)$ is observed, while $\Delta G(^*CO)$ is not scaled with either $\Delta G(^*H)$ or $\Delta G(^*HOCO)$. Such observations suggest that the CO production on TM-Pcs could be controlled by the optimal binding mode of *HOCO and *CO. While the weakened *CO adsorption makes its desorption a facile process, *HOCO adsorption concurrently becomes too weak to facilitate the CO₂RR in the cases of Cu- and Zn-Pcs. On the other hand, despite its strong *HOCO binding, Fe-Pc does not readily produce CO due to CO-poisoning originating from the *CO binding being too strong. As such, the highest CO production with Co-Pc can be understood as a result that the isolated Co

center achieves the most stabilized *HOCO and a decent tolerance against CO-poisoning (**Figure 6C**). In an attempt to find a key descriptor for the CO production over the isolated TM environment, experimentally measured J(CO) values of TM-Pcs were correlated with a combined effect of *HOCO and *CO binding. As shown in **Figure 6D**, the free energy difference values between $\Delta G(^*HOCO)$ and $\Delta G(^*CO)$ (*i.e.*, $\Delta G(^*HOCO) - \Delta G(^*CO)$) generally are consistent with the J(CO) trend of TM-Pcs observed in the experiments, thus making the $\Delta G(^*HOCO) - \Delta G(^*CO)$ value a potential descriptor of the CO₂RR activity over TM-Pcs. In this regard, this result supports a rationale that the CO₂RR would be facilitated when the *HOCO intermediate is stabilized under the condition of a facile *CO desorption. The linear scaling of $\Delta G(^*HOCO) - \Delta G(^*CO)$ and J(CO) maintain well over the entire potential ranges in this study (**Figure S18**), supporting the effectiveness of $\Delta G(^*HOCO) - \Delta G(^*CO)$ as a potential descriptor in explaining the selectivity over the TM-N₄ electrocatalytic center.

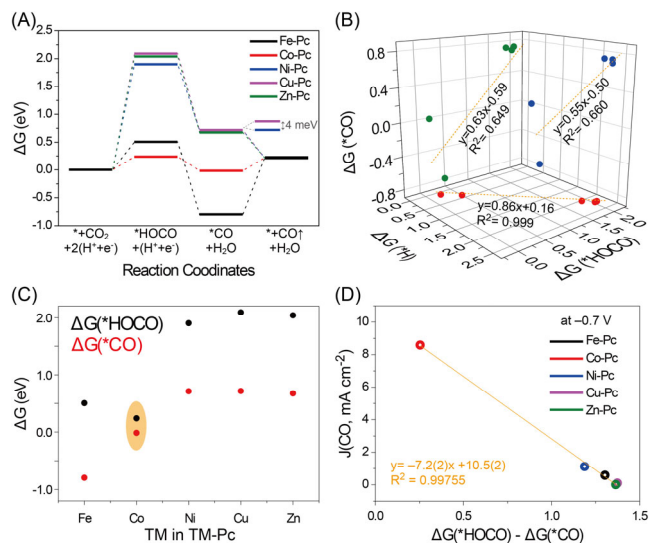


Figure 6. Correlation between experimental results and DFT calculations. (A) Free energy diagram for CO₂RR at $V = 0$ V. (B) A linear scale between the free energy of reaction intermediates over different TM-Pcs. (C) The free energy of *HOCO and *CO with different TM-Pcs. (D) A plot of the free energy difference of *HOCO and *CO adsorptions *versus* J(CO) at -0.7 V.

TM-Pcs provide a unique platform to unambiguously investigate the influence of well-defined isolated TM centers toward catalytic activity and selectivity as opposed to the graphene-based TM-SACs which contain a mix of TM-N_x ($x=0-4$) centers on graphene plane and edges.⁵⁵⁻⁵⁸ In order to clarify the discrepancy of CO₂RR activity and selectivity between TM-Pc and TM-SACs, additional DFT calculations were performed to obtain the density of states

(DOS) of TMs in TM-Pcs. The stronger binding affinity of Fe and Co for the reaction intermediates primarily originates from the higher DOS available at the Fermi level (**Figure S19**). A comparison of calculated DOS of Co in Co-Pc and Co-SAC (*i.e.*, Co-N₄ structure, **Figure S20**) shows the availability of nearly the same number of electronic states at the Fermi level (**Figure S21**). This suggests a similar binding affinity of Co-Pc and Co-SAC for the reaction intermediates as observed in the DFT calculations (**Table S5**). However, it should be noted that Co-Pc and Co-SAC have a difference in the preferred reaction pathways (CO₂RR vs. HER, respectively). Therefore, besides the slightly different TM coordination environment in TM-Pc and TM-SAC, the structural inhomogeneity in the SACs consisting of a mix of TM-N_x active centers potentially facilitates both the CO₂RR and HER, most likely plays a role in the different CO₂RR activity profiles as suggested in recent studies.^{55, 59, 60}

3. Conclusions

In summary, the CO₂RR performance was studied on a series of TM-Pcs (TM = Fe, Co, Ni, Cu, and Zn), which offer a way to unambiguously investigate the intrinsic role of the isolated TM with identical local coordination environments on the activity and selectivity, using a combination of electrochemical evaluations, *in-situ* X-ray analyses, and DFT calculations. It is found that Co-Pc achieves the best CO₂RR activity (FE(CO) and J(CO)) among all TM-Pcs in the present study due to its optimal binding strength of *HOCO and *CO on the catalytic active TM centers. The DFT calculated binding energies indicate that the binding affinity of TM centers in TM-Pc for the reaction intermediates determines the CO₂RR activity. Furthermore, the observed linear correlation between the DFT-computed free energy difference of $\Delta G(*\text{HOCO}) - \Delta G(*\text{CO})$ and experimentally observed J(CO) suggests that $\Delta G(*\text{HOCO}) - \Delta G(*\text{CO})$ can be used as a potential descriptor of the CO₂RR activity to help design improved TM-Pc and well-defined SAC catalysts.

4. Methods

4.1. Chemicals

Potassium carbonate (K₂CO₃), a series of transition metal-based phthalocyanine (TM-Pc, TM = Fe, Co, Ni, Cu, and Zn), Nafion solution (5 wt%), Nafion 117 membrane, and isopropanol were purchased from Sigma Aldrich. Carbon paper (10 wt% water proofed) was purchased from Fuel Cell Store. Polyimide capillary tubing was purchased from Core-Parmer. Vulcan XC-72 was purchased from Cabot Corporation.

4.2. Preparation of Electrolytes

All reagents were used without purification. Unless otherwise noted, all of the chemicals were purchased from Sigma Aldrich. 0.5M potassium bicarbonate (KHCO₃) in deionized (DI) water was prepared by bubbling CO₂ gas for

12 h into 0.25M potassium carbonate (K₂CO₃) in DI water. The pH values of these solutions were ~7.2 after the full saturation of CO₂.

4.3. Preparation and Characterization of Catalysts

The series of TM-Pcs were intimately mixed with Vulcan XC-72 carbon powder with a mass ratio of 1:1 to compensate the insufficient electrical conductivity of TM-Pc and help the dispersion of TM-Pc. For the preparation of working electrode, 2 mg of the TM-Pc/C catalysts was first dispersed in the mixture (1 mL) of deionized water and isopropanol solution (*v:v=1:1*) containing 0.05 wt% Nafion (20 μ L). After sonication for 20 min, the ink was dropped on water-proofed carbon paper (Toray, TGP-H-060, 10%) and dried overnight. The catalyst ink was cast on both sides of carbon paper to avoid, if any, the contribution of carbon paper on CO₂RR. The areal mass loading of TM-Pc/C was fixed to be 400 μ g cm⁻².

Synchrotron powder X-ray diffraction data was obtained on the 17-BM at Advanced Photon Source (APS) with a detector distance of 800 mm. The catalysts were loaded into capillary tubes (Polyimide, Cole-Parmer, Model: 95820-90), followed by plugging the ends with rubber clay. A Perkin-Elmer amorphous silicon detector was used. The ring patterns were integrated and converted into diffraction patterns using GSAS-II software.⁶¹ Typical morphology/structure observation and elemental mapping of the samples were performed using high angle annular dark field scanning TEM (HAADF-STEM, JEM-ARM200F, double spherical aberration correctors, 200 kV).

4.4. Electrochemical Measurements

Saturated calomel electrode (SCE) and graphite paper were used as reference and counter electrodes, respectively. The CO₂-saturated 0.5M KHCO₃ were utilized as electrolyte. The reference electrode was calibrated before each electrolysis experiment by using a reversible hydrogen electrode (HydroFlex, ET-070, EDAQ).

A typical air-tight two-compartment cell was filled with CO₂-saturated solution. In each compartment, 18 mL of the electrolyte was filled, leaving ~13 mL of empty head space. A proton exchange membrane (Nafion 117) was used to separate chambers containing catholyte and anolyte. The working electrode was placed together with reference electrode in the air-tight cathodic chamber. The counter electrode was located in the anodic chamber.

The electrochemical measurements were carried out using a Biologic potentiostat (SP-150). The applied potential (V) was controlled by *i*R-compensation (80%) and converted to the reversible hydrogen electrode by using the following Equation (1)^{13, 62, 63}:

$$V_{\text{RHE}} = V_{\text{SCE}} + 0.248 + 0.059 \times \text{pH} \quad (1)$$

Thus, all of the potentials (V) in this study are written based on the reversible hydrogen electrode potential.

In order to achieve stable electrochemical CO₂ reduction data, 5 cyclic voltammetry (CV) scans were first performed with a scan rate of 100 mV s⁻¹ from 0.1 to 1.2 V. After that,

the electrode was scanned from 0 to -1.1 V by using the linear scanning voltammetry (LSV) technique for three scans. Then, five additional CV scans were used to get rid of the products from the previous LSV step under CO₂ purging. After the additional CO₂ bubbling for 10 min, the electrochemical CO₂RR performance was evaluated by using the chronoamperometry method at each constant potential for different periods. The vigorous magnetic stirring was applied during the electrolysis to help mitigate the mass transport limitation of dissolved CO₂ in the electrolyte. The gaseous product distribution was analyzed by using gas chromatography (GC, Agilent, Agilent 7890B). Once each CA was completed for CO₂RR, 100 μ L of gas in the empty head space at the cathodic compartment was collected and injected into GC to quantify the gaseous products. The liquid product was analyzed by using high performance liquid chromatography (HPLC, 1260 Infinity-II equipped with Hi-Plex H columns, Agilent).

ASSOCIATED CONTENT

Supporting Information. Additional experimental details, XANES profiles, EXAFS fitting results, additional electrochemical data, additional DFT calculation results, transmission electron microscopy images, and additional tables. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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Author Contributions

The manuscript was written with contributions of all authors. All authors have given approval to the final version of the manuscript. *These authors contributed equally.

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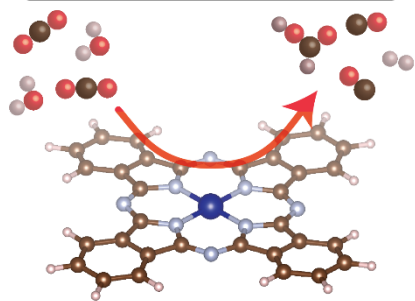
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*Electroreduction of CO₂ over
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TOC Graphic
